

Procedure: Cystoscopy

Code the main condition as [QA06](#) *Follow-up examination after treatment for malignant neoplasms*. Option 1: Code [QC40.5](#) *Personal history of malignant neoplasm of urinary tract*. Option 2: Code [QC40.5](#) *Personal history of malignant neoplasm of urinary tract* as an other condition and postcoordinate the stem code [2C94.Z](#) to specify the personal history is related to bladder cancer. Refer to Section [2.23.16 Coding 'History of and 'Family history of'](#). Main condition: [QA06](#) Other condition: Option 1: [QC40.5](#); Option 2: [QC40.5/2C94.Z](#)

Malignant neoplasms of independent, multiple primary sites

The stem code for Malignant neoplasms of independent (primary) multiple sites should be coded as the main condition when the health care practitioner records as the main condition two or more independent primary malignant neoplasms, none of which predominates. Then, optionally, additional codes to identify the individual neoplasms may be coded as other conditions to identify the individual primary malignant neoplasms recorded by the health care practitioner. Extension codes may be added to each primary malignant neoplasm stem code to identify additional detail of the histopathology and the site.

Example 1:

The documentation states that the patient has carcinomatosis of the peritoneum from an unknown primary neoplasm.

Main condition: Carcinomatosis of peritoneum

Code the main condition as [2D91](#) *Malignant neoplasm metastasis in peritoneum*. Code as another condition [[2D4Z](#) *Unspecified malignant neoplasms of unspecified sites*]. Main condition: [2D91](#) Other condition: [2D4Z](#)

Example 2:

Main condition: Multiple myeloma

Other conditions: Primary adenocarcinoma of prostate

Code the main condition as [[2A83.1](#) *Plasma cell myeloma*]. Code as an other condition [[2C82.0](#) *Adenocarcinoma of prostate*]. Main condition: [2A83.1](#) Other condition: [2C82.0](#)

Unspecified malignant neoplasms of unspecified sites

This code should be used only when the health care practitioner has clearly recorded the neoplasm as an unknown primary site or as an unspecified malignancy, assumed primary.

Malignant neoplasm metastases, unspecified site

This code should be used as the main condition only when the malignancy is described as 'disseminated metastases' or 'metastatic carcinoma' (or other similar terms as described in the inclusion list of the code) but the specific sites are not documented.

2.23.21.3 Chapter 3: Diseases of the blood or blood-forming organs

Certain conditions classifiable to this chapter may result from drugs or other external causes. Codes from Chapter 23 'External causes of morbidity and mortality' may be used as optional additional codes.

Example 1:

Patient who is on long-term treatment with the drug trimethoprim is admitted and treated for trimethoprim-induced folate deficiency anaemia. Main condition: Trimethoprim-induced folate deficiency anaemia Code the main condition as [\[3A02.4 Drug-induced folate deficiency anaemia\]](#) and postcoordinate with the external cause code [\[PL00 Drugs medicaments and biological substances associated with injury or harm in therapeutic use\]](#) and the external cause code [\[PL13.2 Drug-related injury or harm in context of correct administration or dosage, as mode of injury or harm\]](#). The extension code [\[XM7NY9 Trimethoprim\]](#), may be added optionally to identify the drug. Main condition cluster: [3A02.4 /PL00&XM7NY9 /PL13.2](#)

2.23.21.4 Chapter 5: Endocrine, nutritional or metabolic diseases

Certain conditions classifiable to this chapter may result from drugs or other external causes. Codes from Chapter 23 'External causes of morbidity and mortality' may be used as optional additional codes.

Diabetes mellitus

When the health care practitioner has documented a condition as due to diabetes mellitus, postcoordinate the condition and the diabetes mellitus stem codes. If more than one condition is documented as being due to diabetes mellitus, each distinct clinical concept (each diabetes caused condition) is coded on its own and postcoordinated with the diabetes mellitus stem code even though it means repeating the diabetes code in each cluster. (Refer to Example 2 below).

Example 1:

Main condition: Chronic kidney failure due to type 2 diabetes mellitus. Chronic kidney failure is documented as due to diabetes mellitus; therefore, code to [GB61.Z Chronic kidney disease, stage unspecified](#) and postcoordinate with the stem code [5A11 Type 2 diabetes mellitus](#). Main condition cluster: [GB61.Z/5A11](#)

Example 2:

Main condition: Type 1 diabetes with diabetic nephropathy Other condition: Diabetic cataract Code the main condition as [5A10 Type 1 diabetes mellitus](#) postcoordinated with the stem code [GB61.Z Chronic kidney disease, stage unspecified](#). Code as an other condition [9B10.21 Diabetic cataract](#) postcoordinated with the stem code [5A10 Type 1 diabetes mellitus](#). Main condition cluster: [5A10/GB61.Z](#) Other condition cluster: [9B10.21/5A10](#)

Carcinoid syndrome

This code is not to be used as the preferred code for main condition if a carcinoid neoplasm is recorded, unless the episode of care was directed predominantly at the endocrine syndrome itself.

2.23.21.5 Chapter 6: Mental, behavioural or neurodevelopmental disorders

Dementia

Always code the underlying aetiology, if documented.

2.23.21.6 Chapter 8: Diseases of the nervous system

Certain conditions classifiable to this chapter may result from the effects of drugs or other external causes. Codes from Chapter 23 'External causes of morbidity and mortality' may be used as optional additional codes.

Late effect of cerebrovascular disease

These codes are not to be used as the preferred code for the 'main condition' if the nature of the residual condition is recorded. Refer to Section [2.23.18 Coding of conditions documented as sequela \(late effect\)](#).

Paralytic symptoms

These codes are not to be used as the preferred code for the main condition if a current cause is recorded, unless the episode of care was mainly for the paralysis itself.

Example 1:

Patient is admitted with left side hemiplegia and following investigation determined to be due to acute ischaemic stroke. Main condition: Acute ischaemic stroke with hemiplegia. Code the main condition as [8B11.5Z Cerebral ischaemic stroke, unspecified](#) and postcoordinate the stem code [MB53.Z Hemiplegia, unspecified](#). An optional extension code to specify [XK8G Left](#) may be added. Main condition cluster: [8B11.5Z Cerebral ischaemic stroke, unspecified](#)/[MB53.Z](#)&[XK8G](#)

Example 2:

Patient is admitted for rehabilitation training for paralysis of left leg resulting from cerebral infarction three years ago. Main condition: Paralysis of left leg Code the main condition as [MB55.Z Monoplegia of lower extremity, unspecified](#) and an optional extension code to specify [XK8G Left](#) may be added. Postcoordinate the stem code [8B25.0 Late effects of cerebral ischemic stroke](#) Main condition cluster: [MB55.Z](#)&[XK8G](#)/[8B25.0](#)

2.23.21.7 Chapter 9: Diseases of the visual system

Vision impairment including blindness

These codes are not to be used as the preferred code for the main condition if the cause is recorded, unless the episode of care was mainly for the blindness itself.

2.23.21.8 Chapter 10: Diseases of the ear or mastoid process

Acquired hearing impairment

These codes are not to be used alone if the cause is recorded, unless the episode of care was mainly for the hearing loss itself.

2.23.21.9 Chapter 11: Diseases of the circulatory system

Secondary hypertension

This code is not to be used as the preferred code for the main condition if the cause is recorded. When coding to the cause, secondary hypertension is used as an additional code (in a cluster) to indicate that this manifestation has been relevant in the context of treatment.

2.23.21.10 Chapter 15: Diseases of the musculoskeletal system or connective tissue

Many musculoskeletal conditions are treated without knowing the underlying disease. In such cases only the musculoskeletal condition is coded.

2.23.21.11 Chapter 18: Pregnancy, childbirth or the puerperium

JA05 *Complications following abortion, ectopic or molar pregnancy*

These codes are not to be used as the preferred code for the main condition, except where a new episode of care is solely for treatment of a complication, e.g. a current complication of a previous abortion. These codes may be used as an optional additional code with 'Abortive outcome of pregnancy' codes to identify associated complications and to give fuller details of the complication.

Example 1:

Main condition: Ruptured tubal pregnancy causing shock

Other conditions: -

Specialty: Gynaecology

Code the main condition as [JA01.1](#) *Tubal pregnancy* and since the shock is documented as a complication of the tubal pregnancy, postcoordinate [JA05.3](#) *Shock following abortion, ectopic or molar pregnancy*.

Main condition cluster: [JA01.1](#)/[JA05.3](#)

Example 2:

Patient is diagnosed with endometritis following a spontaneous abortion that was diagnosed and treated at a previous episode of care.

Main condition: Endometritis following spontaneous abortion

Specialty: Gynaecology

This example represents a new episode of care solely for treatment of a current complication of a previous spontaneous abortion; therefore, code the main condition as [JA05.0](#) *Genital tract or pelvic infection following abortion, ectopic or molar pregnancy*. No other code is required since the abortion took place during a previous episode of care.

Main condition: [JA05.0](#)

Delivery

Use of these codes to describe the 'main condition' should be limited to cases where the only information recorded is a statement of delivery or the method of delivery. These codes may be used as additional codes to indicate a method or type of delivery where no separate data item or procedural classification is being used for this purpose.

Example 3:

Patient is admitted in labour and delivers a healthy newborn without complication.

Main condition: Normal delivery

Other conditions: -

Procedure: Spontaneous vaginal delivery

Code [JB20.Z](#) *Single spontaneous delivery, unspecified* as 'main condition' and postcoordinate [QA46.0](#) *Single live birth*.

Main condition cluster: [JB20.Z](#)/[QA46.0](#)

Example 4:

Patient who has a history of previous Caesarean section is admitted for in labour. A trial of labour is unsuccessful for vaginal delivery due to arrested active phase and unplanned repeat Caesarean section is performed.

Main condition: Failed trial of labour, unspecified

Other conditions: Secondary uterine inertia

Procedure: Caesarean section

Code [JB0D.8](#) *Failed trial of labour, unspecified* as the 'main condition' and postcoordinate [[JB02.1](#) *Secondary uterine inertia*] because the health care practitioner has documented the cause of the failed trial of labour. Code [JB22.1](#) *Delivery by emergency caesarean section* as an other condition to indicate the method of delivery, and also code the outcome of delivery [[QA46.0](#) *Single live birth*].

Main condition cluster: [JB0D.8](#)/[JB02.1](#)

Other condition cluster: [JB22.1](#)/[QA46.0](#)

Example 5:

Patient who is known to have a twin pregnancy is admitted in labour and delivers two healthy newborns.

Main condition: Multiple delivery, all spontaneous

Other conditions: Twins, both liveborn

Procedure: Spontaneous delivery

Code [JB24.0](#) *Multiple delivery, all spontaneous*/[QA46.2](#)

Main condition cluster: [JB24.0](#)/[QA46.2](#)

Example 6:

Patient is admitted in labour at 38 weeks gestation. On examination, no fetal heart rate could be detected.

Main condition: Maternal care for fetal death

Other conditions: -

Procedure: Spontaneous delivery

Code to [JA86.3](#) *Maternal care for intrauterine death* and postcoordinate [[XT6G](#) *Duration of pregnancy more than 36 completed weeks*]. Code as an other condition [JB20.Z](#) *Single spontaneous delivery, unspecified* to indicate the method of delivery.

Main condition: [JA86.3](#) & [XT6G](#)

Other condition: [JB20.Z](#)

Certain maternal diseases classifiable elsewhere but complicating pregnancy, childbirth or the puerperium

The subcategories provided should be used as 'main condition' codes in preference to categories outside Chapter 18 'Pregnancy, childbirth or the puerperium' when the conditions being classified have been indicated by the health care practitioner to have complicated the pregnant state, to have been aggravated by the pregnancy, or to have been the reason for obstetric care. The pertinent codes from other chapters may be used as optional additional codes to allow specification of the condition. Postcoordination applies when the additional code to specify the condition is coded.

Example 7:

Patient is admitted at 28 weeks gestation with toxoplasmosis.

Main condition: Toxoplasmosis

Other conditions: Pregnancy undelivered

Code [J63.6Z](#) *Protozoal diseases complicating pregnancy, childbirth or the puerperium, unspecified* as the main condition and optionally, code [1F57.Z](#) *Toxoplasmosis, unspecified* to identify the specific protozoal disease that is complicating the pregnancy. When the additional code to identify the specific complication is coded, then postcoordination applies because [1F57.Z](#) is adding additional detail/specificity to the stem code [J63.6Z](#).

Main condition cluster: [J63.6Z/1F57.Z](#)

2.23.21.12 Chapter 21: Symptoms, signs or clinical findings, not elsewhere classified

Categories from this chapter should not be used as ‘main condition’ codes unless the symptom, sign or clinical finding was clearly the main condition treated or investigated during an episode of care. Codes from this chapter should not be assigned where an explanatory diagnosis is determined during the episode of care.

2.23.21.13 Chapter 22: Injury, poisoning or certain other consequences of external causes

Where multiple injuries are recorded and no one of these has been selected as the ‘main condition’, code to one of the categories provided for statements of multiple injuries of:

- same type to the same body region;
- different types to the same body region; and
- same type to different body regions

and postcoordinate the stem codes that describe each individual injury.

Note the following exceptions:

- for internal injuries recorded with superficial injuries and/or open wounds only, code to internal injuries as the ‘main condition’;
- for fractures of skull and facial bones with associated intracranial injury, code to the intracranial injury as the ‘main condition’;
- for intracranial haemorrhage recorded with other injuries to the head only, code to intracranial haemorrhage as the ‘main condition’;
- for fractures recorded with open wounds of the same location only, code to fracture as the ‘main condition’.

When the multiple injury categories are used, codes for any individual injuries listed are used as additional codes in the same cluster.

Example 1:

Patient suffered injuries to the bladder and urethra following an assault. Main condition: Injury to bladder and urethra Other conditions: - Code as main condition [NB92.8](#) *Injury of multiple pelvic organs* and postcoordinate the stem codes for [NB92.2Z](#) *Injury of bladder, unspecified* and [NB92.3Z](#) *Injury of urethra, unspecified* as these codes are adding additional detail/specificity to [NB92.8](#). Main condition cluster: [NB92.8/NB92.2Z](#)

Example 2:

Patient, who was the driver of a motorcycle, lost control on the highway and crashed. Investigations revealed open intracranial wound with cerebellar haemorrhage. Main condition: Open intracranial

wound with cerebellar haemorrhage Other conditions: - Code the main condition [NA07.82 Traumatic haemorrhage in cerebellum](#). Code the other condition [NA07.Y Other specified intracranial injury](#).

2.23.21.14 Chapter 23: External causes of morbidity or mortality

These codes are not to be used as 'main condition' codes. They are intended for use as additional codes to identify the external cause of conditions classified in Chapter 22 and may also be used as optional additional codes with conditions classified in any other chapter, but which have an external cause.

2.23.21.15 Chapter 24: Factors influencing health status or contact with health services

There are some health care episodes that are not related to the treatment of or investigation for current illness or injury (e.g. monitoring of previously-treated conditions, immunisation visits, seeking of health-related advice). In such circumstances, a code for the main condition can potentially be found in Chapter 24 'Factors influencing health status and contact with health services'.

2.23.22 Traditional Medicine Conditions - Module 1 (TM1)

Traditional Medicine (TM) is an integral part of health services provided in many countries. International standardisation by including Traditional Medicine conditions within the ICD allows for measuring, counting, comparing, formulating questions and monitoring its use over time. ICD-11's supplementary chapter on Traditional Medicine disorders and patterns (TM1) is designed to be used in conjunction with the Western Medicine concepts of ICD Chapters 01-25.

As with other ICD chapters, the TM1 chapter is a tool for classifying, diagnosing, counting, communicating and comparing TM conditions, it will also assist research and evaluation to assess the safety and efficacy of TM. This chapter not judging or endorsing TM practice or the efficacy of any TM intervention.

2.23.23 Use in Traditional Medicine

Reporting at regional, national and international levels:

- Counting episodes of care for Traditional Medicine disorders and/or patterns in the same way as for Western Medicine diseases for morbidity data reporting purposes
- Counting episodes of care by Traditional Medicine practitioners who may use a combination of Western Medicine and Traditional Medicine terminology
- Describing and quantifying utilisation of Traditional Medicine services and reasons for encounter
- Monitoring use of resources for Traditional Medicine services
- Standardising descriptions of disorders and patterns among TM clinicians, practitioners and coders

Research:

- Facilitating evidence-based research on safety and efficacy of Traditional Medicine interventions
- Allowing clinical research within the Traditional Medicine framework and integrating Western Medicine with Traditional Medicine
- Understanding interrelationships between Western Medicine diseases and Traditional Medicine disorders and patterns
- Studying treatment patterns and outcomes for specific disorders and patterns using ICD-11 in conjunction with country specific procedure classifications and the TM component of the International Classification of Health Interventions (ICHI)

Casemix reimbursement and insurance:

- There are precedents in China, Japan, and Korea for use of existing Traditional Medicine classifications (with or without Western Medicine concepts) for reimbursements to hospitals and for insurance claims and for clinical costing measures.
- Incorporating Traditional Medicine as a chapter of ICD-11 allows much greater scope for describing patient conditions (diseases, disorders (TM1) and patterns (TM1) across the Western Medicine and TM1 chapters) as well as complications and comorbidities.

Quality and safety of care:

- Standardising use of codes reflecting quality and safety of care between Western Medicine diseases and TM1 disorders will allow TM practitioners to interpret data from ICD-11 on quality, safety, and efficacy of care.

Education:

- Educating TM practitioners in regard to standardisation of diagnosis
- Educating TM clinicians and coders in application and interpretation of ICD-11 data.

Standardising terminology for use in electronic health records:

- To enable more consistent and efficient recording and extraction of data
- To allow computer assisted coding of TM1 disorders and patterns

2.23.24 Coding instructions for Traditional Medicine conditions - Module 1 (TM1)

Codes from the Traditional Medicine chapter can be used across settings (hospital inpatient or ambulatory care in hospital or community) but must not be used for reporting cause of death. When coding in primary care, disorders and patterns may not be fully developed so that it may be more feasible to identify reason for encounter rather than main condition and associated conditions.

General principles:

- Consult all parts of the patient record including discharge summary, history, physical examination, investigations, laboratory data, treatments and final diagnoses.
- Coding should relate to reasons for treatment during this episode and need not describe the whole patient's lifetime history unless a past condition affects current care.
- Be as specific and explicit as possible, using codes to represent aetiology, pathology and manifestations of TM condition.
- Use codes from relevant chapters of the ICD to match the clinical disorders noted on the patient record.
- Code threatened TM conditions (i.e. those not well defined or not manifest).

The inclusion of the supplementary chapter on Traditional Medicine conditions in ICD-11 aims to foster integration of TM into the health system. Hence, it is recommended that - wherever possible - TM1 codes should be used in conjunction with those from other ICD-11 chapters (Chapter 01-25) to enable comparison of Western medicine diagnosis and Traditional Medicine diagnosis.

The following generic coding scheme and alternative coding options can be used:

1. Read the patient summary and medical record.
2. Select WM diagnosis/diagnoses, TM1 disorder(s) (TM1), and/or pattern(s) (TM1) to be coded.

Options	Examples	ICD-11 Coding Examples
a. WM diagnosis alone	Asthma	CA23.32
b. WM diagnosis with TM1 pattern	Asthma	CA23.32
	Turbid phlegm accumulation in the lung pattern (TM1)	SF86
c. WM diagnosis with TM1 disorder	Asthma	CA23.32
	Wheezing disorder (TM1)	SF81

You may choose more than one disorder (TM1) and more than one pattern (TM1) from the TM chapter.

3. Consult the electronic Coding Tool or relevant Alphabetic Indexes for WM and TM1 entries
4. Go to tabular list for the relevant code. Take note of inclusions and exclusion notes and textual descriptions.
5. Assign the appropriate code and follow any specific guidelines for that code.

Applying the above listed generic coding scheme and alternative coding options for a given clinical picture must take into account the regulatory context, coding practice and setting specific requirements at country level. Additional coding conventions may be developed in response to country specific information needs (e.g. coding of main condition, use of

extension codes, coding of neoplasms and injury, chronic and complicated conditions, sub-clinical or constitutional complaints, external cause of injury and adverse reaction).

2.24 General statistical recommendations

2.24.1 Data quality

To ensure high quality, processes for monitoring the quality of the coded data need to be implemented. This is referred to as Quality Assurance. On the following pages you will find some suggestions relating to the application of Quality Assurance for mortality and morbidity statistics. As a basic principle, those responsible for the collection and analysis of data should be involved in the development of the protocols for the processing and coding of the data, and in determination of the other data items to be cross-tabulated with them. Collecting quality data requires a clearly designed workflow (from reporting to coding to analysis), and adequate training of all involved parties. In particular, all participants need to understand the process and their part in it. The basic stages of the workflow include:

1. Documentation – This is where the information starts. Identifying a condition and then documenting it on a death certificate, in a medical record or on another medical form need to be carried out completely and accurately, using the best possible evidence. For this reason, this part of the workflow, should generally be carried out by a well-trained physician.
2. Reporting – Documented information should be reported in the format specified by the responsible agencies. There may be differences in the required reporting structure, timeframe and obligations, depending on the country and/or data collection circumstances.
3. Verification– Where unclear information, incomplete or illogical statements are reported, a feedback loop and return of queries to the certifiers and documenters should be used.
4. Grouping and analysis – Both serve to aggregate data in ways that are determined by the diverse use cases. Rules and constraints should be clearly understood and communicated when analysis is undertaken and results are reported and used.

2.24.2 Specificity versus ill-defined codes

Reported information should be coded to the highest level of detail possible. In some instances, the details available are sparse, incompletely documented or only trivial details are reported. Though the ICD provides categories for these cases, such non-specific information does not allow understanding of epidemiological patterns and/or utilisation of health systems nor does it support prevention of disease.

2.24.3 Problems of a small population

Population size is one of the factors that need to be considered when the health status of a population is assessed by means of mortality or morbidity data. In countries with small populations, the annual numbers of events in many categories will be very small and may fluctuate from year to year, especially when disaggregated by recommended age and sex. These problems can be alleviated through one or more of the following measures:

- use or presentation of broad groupings of ICD rubrics, such as chapters
- aggregation of data over a longer time period, e.g. to take the preceding two years of data together with those for the current year and produce a 'moving average' figure
- using the broadest possible age groupings

The recommendations that apply for small national populations also hold true, in general, for reporting the data relating to subnational segments of larger populations. Investigations of health issues in population subgroups have to take into consideration the effect of the size of each of the subgroups on the type of analysis used. This need is generally recognised when dealing with sample surveys, but often overlooked when the investigation concerns the health problems of special groups in a national population.

2.24.4 'Empty cells' and cells with low frequencies

Regardless of the list of causes being used, it may be found that no cases for one or more listed cause(s) occur in certain cells in a statistical table. Where there are many empty lines in a table, it is worth considering the omission of such lines from a published table. When only the occasional case of a disease occurs in a country, the line can be regularly omitted from the published statistical table and a footnote added to indicate either that there were no cases in the reported timeframe or, when sporadic cases do occur, in which cell the case would have appeared. For cells with very low frequencies, especially those relating to diseases that would not be expected to occur, it is important to establish that the cases existed and did not result from a coding or processing error. This should be carried out as part of the general quality control of the data.

2.24.5 Precautions needed when tabulation lists include subtotals

It may not always be apparent to those processing the data that some of the items in the tabulation lists are in fact subtotals or earlier reported categories. These items may include titles of blocks and titles of four-character categories or the items for chapter titles (in the condensed versions of the mortality tabulation lists). These entries should be ignored when totals are calculated, otherwise cases may be counted more than once.

2.24.6 Ethical Aspects

Confidentiality refers to the obligation to not disclose information (data to third parties where that information has originally been delivered in confidence. This duty was codified in the Hippocratic Oath in the 4th century BCE and is still one of the core principles of medical ethics. Any information that might allow the identification of a specific person should only be viewed by people who are authorised to do so. Authorisation means that a person is legally permitted to access the information. e.g. medical staff, coroners and coders are all people who may be authorised to see sensitive information.

Generally, the only way for confidential information to become publicly accessible is through legislation, statutes and regulations. Sometimes confidential information can become public record after a certain period of time. For instance, in some regions of the world the passage of time can render death certificates matters of public record and, therefore, the requirement for them to remain confidential no longer applies.

The authorised supplier of confidential information must verify that the requesting person is an authorised user and determine their level of authorisation. The supplier must be aware of the level of information that can be made available to the authorised user and take appropriate steps to guard against unauthorised disclosure. The authorised user must not attempt to gain access to information which they are not authorised to obtain. Additionally, the user must also guard against unauthorised access to the information that has been supplied to them. This means that users must secure the confidential information and any recordings of that information in a way that prevents unauthorised access. The information must only be used for appropriate purposes and must be returned as required. National legal frameworks, state and local regulations, and institutional guidelines provide specific rules and information regarding how to maintain confidentiality.

2.24.7 Avoidance of Potential Harm

Direct and serious harm can result from a breach of confidentiality. For example, the disclosure of sensitive information can potentially lead to stigma and discrimination against an individual. Conversely, greater harm can result from maintaining confidentiality than from not doing so. Some circumstances may require a judgement that involves balancing the harms to or disclosure against the interests of, the patient, deceased person and other relevant parties. A person may suffer 'harm' physically, socially or psychologically as a result of a breach of confidentiality. A confidential diagnosis that is breached makes the patient lose faith or trust in the clinician, and the patient may then suffer abuse from another person or suffer the stigma associated with certain conditions. In other circumstances the nondisclosure of one person's confidential information may result in another individual or a community being at risk of developing a harmful condition or being exposed to a harmful situation.

This is a difficult concept and one that must be approached in a thoughtful way. As previously mentioned, there are times when it is justifiable to give others confidential information, such as when reporting communicable disease incidence. In such cases the reporting of confidential information is usually allowed. This is an example of where the nondisclosure of a disease could result in major harm to others.

If it is necessary to disclose information, it is preferable to contact the relevant person and let them know about the need to do so. In some cases, this may not be possible or appropriate, and users should be guided by legal and institutional guidelines. The harms that can be caused by disclosure of certain information must be considered. Some information that can be particularly sensitive includes tests for genetic disorders and diseases, incidence of communicable diseases, and HIV test results. Sometimes there are special requests for confidentiality that may require increased levels of confidentiality assurances. These special requests cannot supersede legal requirements for disclosure but should be respected when possible.

2.24.8 Security of Privacy – Confidentiality

Privacy relates to protecting an individual's control over what personal information and decisions may or may not be shared with others. For instance, when a physician examines or speaks with a patient it is usually done in a non-public area so that the information given to the physician by the patient cannot be heard by anyone else. It also enables the physician to

give a patient their diagnosis in private. Data are shared with consent by the patient or where required by law or legislation. Security of privacy and confidentiality of health (and other) data are usually addressed by national laws and regulations.

2.25 Recommendations in relation to statistical tables for international comparison

The following recommendations aim to standardise the presentation of coded data to allow national and international comparisons between different countries or regions.

2.25.1 The recommended Special tabulation lists

There are standard ways of listing causes coded according to the ICD, and there are formal recommendations concerning lists for tabulation that allow for national and international comparisons. The hierarchical structure of the ICD allows considerable flexibility for other possible tabulations. For mortality, the ICD includes special tabulation lists which are intended for circumstances in which the four-character list is too detailed, and are designed so that international comparison of significant diseases and groups of diseases is not compromised by different groupings used in different countries.

It should be noted that the special tabulation lists are designed for the aggregation and reporting of coded data. They are not for coding purposes.

The special tabulation lists are:

- List 1 Mortality tabulation list
- List 2 Morbidity tabulation list
- List 3 International Shortlist for Hospital Morbidity Tabulation (ISHMT)
- List 4 Infectious diseases by agent condensed list
- List 5 Sustainable Development Goals (SDGs)
- List 6 WHO Verbal autopsy list

The mortality list is based on the codes assigned for the underlying cause of death.

Morbidity list do not include codes from traditional medicine conditions.

Use of prefixes to identify the special tabulation lists

Use of the numerical prefixes prevents confusion between the special tabulation lists, as the ICD four-character codes have a letter in the second position. Where an adapted list is used for national or sub-national purposes, an alternative identifying prefix should be used.

The special tabulation list for mortality

The mortality tabulation list provides items for each ICD chapter and also, within most chapters, identify the items with residual items entitled 'Remainder of...' that complete the coverage of the respective chapter. The list covers the full range of ICD four-character codes applicable for mortality, required for reporting and publication purposes.

Locally designed lists for mortality

For most countries, the mortality tabulation list provides enough information on the most important diseases and external causes of death. They also facilitate comparison over time and observation of shifts in the relative frequencies as health programmes take effect, of e.g. infectious diseases and degenerative diseases. It permits comparison between sub-national areas and population sub-groups. In addition, it makes meaningful international comparisons of causes of death possible.

Lists similar to the special tabulation list can also be designed for local use. The ICD rubrics in such lists can be selected and grouped in any way. Special lists may be needed, for example, for monitoring progress, in terms of mortality and also morbidity, of many local health programmes. When adapting the special tabulation lists to national requirements, or when a tabulation list is being devised for a new or special project, a trial run is helpful by counting the number of cases for each four-character category. In such way, it can be determined which conditions are appropriate for broad grouping, and where subcategories are necessary.

Where a local list is constructed, the key to the condensed categories should contain the same four- (or five-) character codes of the core classification.

The special tabulation list for morbidity

The morbidity special tabulation list is intended as a basis for national lists and for intra- and inter-country comparison. National lists can be constructed by either condensing or expanding the core classification as appropriate. The list is suitable for data on inpatient care and, with suitable adaptation (notably through aggregation of some items and expansion of items relating to Chapter 21 'Symptoms, signs or clinical findings, not elsewhere classified' and Chapter 24 'Factors influencing health status and contact with health services') for information from other sources, such as ambulatory care and surveys.

When a local list is constructed, the key to the condensed categories should contain the four (or five) character codes of the core classification. The list has been designed for international comparisons of hospital morbidity statistics. This concise list allows for comparison of hospital activity, independent of health systems. The conditions have been selected in a way that they can always be treated in an admission of at least 24 hours. If, after examination of the frequencies of the ICD four-character rubrics, it is necessary to expand the list, some of the items within ICD categories can be subdivided according to the core classification or even to the five-character level. If the recommended list is too detailed or if a shorter list is required, selection can be made based on national or local health concerns. Depending on a country's 'epidemiological profile', categories may be combined to shorten the list.

2.25.2 International morbidity reporting

International morbidity reporting and comparison of data among different countries requires internationally agreed definitions of:

- inpatient, recoding of day-patients, outpatient
- hospital
- treatment episode
- reason for encounter used instead of diagnosis

See Section [1.4](#) for more detail.

2.25.2.1 Minimum data set and markup for postcoordination

A minimum data set suitable for international comparison would include age, sex, main diagnosis, (reason for admission after assessment at the end of the stay), and health sector (hospital, practitioner, other), is ideally accompanied by the definitions in place for the variables mentioned above, and the main intervention. In an extended data set, ideally, additional diagnoses and interventions are reported in separate data fields. The markup for international reporting of postcoordinated codes in clusters will follow the specifications below:

- a slash '/' separates 2 stem codes
- an '&' links stem codes with extension codes
- a cluster may consist of a single code or One condition with additional detail in one cluster
- stem code & extension code & extension code

Two unrelated conditions will have two clusters:

- stem code - stem code

Two clusters with multiple codes:

- stem code & extension code/stem code & extension code & extension code

Example 1:

Diabetes mellitus / Diabetic retinopathy

Example 2:

Multiple fractures of forearm / fracture of shaft of ulna & compound fracture / fracture of shaft of radius & compound fracture /external cause code

2.25.3 Presentation of statistical tables

The degree of detail in cross-classification by cause, sex, age, and geographical area will depend both on the purpose and range of the statistical reportings and on the practical limits to their tabulation. The following patterns, which are designed to promote international compatibility, present standard ways of expressing various characteristics. Where a different classification is used in published tables (e.g. in age-grouping) it should be reducible to one of the recommended groupings.

- (a) Analysis by the International Classification of Diseases should, as appropriate, be in accordance with:

- the detailed list of four-character categories, with or without five or six-character subcategories;
 - the special tabulation list for mortality;
 - the special tabulation list for morbidity.
- (b) Age classification for general purposes:
- under 1 year, 1 to 4 years, 5-year groups from 5 to 84 years, 85 years and over, 95 years and over;
 - under 1 year, 1-4 years, 5-14 years, 15-24 years, 25-34 years, 35-44 years, 45-54 years, 55-64 years, 65-74 years, 75 years and over.
 - under 1 year, 1-14 years, 15-44 years, 45-64 years, 65 years and over.
- (c) Classification by area should, as appropriate, be in accordance with:
- each major civil division;
 - each town or conurbation of 1,000,000 population and over, otherwise the largest town with a population of at least 100,000;
 - a national aggregate of urban areas of 100,000 population and over;
 - a national aggregate of urban areas of less than 100,000 population;
 - a national aggregate of rural areas.

Note 1. Statistics relating to (c) should include the definitions used to determine urban and rural.

Note 2. In countries where medical certification of the cause of death is incomplete or limited to certain areas, figures for deaths not medically certified should be published separately.

2.25.3.1 Tabulation of causes of death

Statistics of causes of death for a defined area should be in accordance with recommendation 'Statistical tables' (a)(1) above, or, if this is not possible, with recommendation 'Statistical tables' (a)(2).

Deaths should preferably be classified by sex and age group as in recommendation 'Statistical tables' (b)(3).

Statistics of causes of deaths for the areas in recommendation 'Statistical tables' (c) should comply with recommendation 'Statistical tables' (a)(2), or if this is not possible, with recommendation 'Statistical tables' (a)(3). They should preferably be tabulated by sex and age group as in recommendation 'Statistical tables' (b)(2).

2.25.3.2 Injury mortality

Injury mortality traditionally distinguishes between injuries that are caused by:

- Interpersonal violence and sexual abuse
- Collective violence including wars, civil insurrections and riots
- Traffic incidents
- Incidents at home, at work, and while participating in sports and other recreational activities

In the context of mortality, the WHO recommends the retention of codes for both main injury and external causes. In places where this is not feasible, the external cause code should be retained. For injury-related deaths, in ICD-11 the external cause code is the single underlying cause of death code, and incorporates the intent, mechanism, and object of the deceased in a single code. Place of occurrence and activity are coded separately.

2.25.4 Standards and reporting requirements for mortality in perinatal and neonatal periods

2.25.4.1 Terms used in perinatal and neonatal mortality

The perinatal period commences at 22 completed weeks of gestation and ends 7 completed days after birth, i.e. includes days 0 – 6 after birth.

The neonatal period commences at birth and ends 28 completed days after birth i.e. includes days 0 – 27 after birth.

Gestational age is the duration of gestation estimated based on the best obstetric estimate of gestation, which is usually expressed in completed weeks with additional days, or in completed days.

The best obstetric estimate of gestation is based on the birth attendant's final estimate of gestation, calculated from time elapsed since the first day of gestation. The first day of gestation is usually determined by:

- 1) the first day of the last menstrual period (LMP) 1 if confirmed by results of early ultrasound scan
- 2) by early ultrasound scan, where LMP and results of early ultrasound scan differ
- 3) by LMP and/or the clinical postnatal estimate of gestational age, where no early ultrasound scan is available.

In cases of assisted reproduction when the date of embryo transfer is known, an offset of 14 days is to be added for calculating gestational age.

Gestational age is counted by calendar days where day zero (Day 0) is used to refer to the first calendar day of gestation and day 1 for the second calendar day. The first day of gestation i.e. day 0 corresponds to gestational age 'zero completed weeks' with zero additional days (gestational age 0+0 completed weeks), and 6 days later would be gestational age 0+6 completed weeks.

The number of completed weeks is calculated as the number of days since the first day of gestation divided by 7, presented as a whole integer plus a remainder. For example, day 8 after gestation is 1+1 completed weeks, and day 252 corresponds to 36+0 completed weeks, and 6 days later (258 days) would be 36+6.

Birthweight is the first weight of the fetus or neonate obtained after birth. For live births, birthweight should preferably be measured within the first hour of life before significant postnatal weight loss has occurred.

Chronological age used in recording deaths in the neonatal period is counted as follows: Day 0 is used to refer to the first 24 hours after birth. Day 1, is the remainder of the 2nd calendar day (date of death=date of birth+1) but outside the 1st 24 hours. Day 2 is the 3rd calendar day (date of death=date of birth+2).

Where it is not feasible to capture information on hours at death, Day 0 should be considered to be the calendar day of birth (date of birth is equal to date of death), and subsequently: Day of death = Date of death - date of birth.

2.25.4.2 Definitions in perinatal and neonatal mortality

Fetal death, spontaneous abortion, stillbirth, live birth, neonatal death

Death in perinatal or neonatal period should be counted based on the time of delivery (i.e. a complete expulsion or extraction from a woman), though it may be diagnosed earlier in utero. Delivery of an embryo or fetus may occur either spontaneously, assisted or by caesarean section (i.e. it includes deliberate interruption of an ongoing pregnancy by medical or surgical means intended to result in a live birth), while it should be distinguished clearly from induced abortion.

- **Fetal death** is death of a fetus prior to its complete expulsion or extraction from a woman, irrespective of the duration of pregnancy.
- **Spontaneous abortion** (also referred to as **miscarriage**) is a spontaneous loss of pregnancy (i.e. embryo or fetus) before 22 completed weeks of gestation.
- **Stillbirth** is the complete expulsion or extraction from a woman of a fetus, following its death prior to the complete expulsion or extraction, at 22 or more completed weeks of gestation.
- **Live birth** is the complete expulsion or extraction from a woman of a fetus, irrespective of the duration of the pregnancy, which, after such separation, shows signs of life.
- **Neonatal death** is a death after live birth which occurs during the neonatal period.

Death of an embryo or fetus may be diagnosed in utero by absence of heart sounds, confirmed by imaging techniques where available, or after the complete expulsion or extraction from a woman by absence of signs of life. Signs of life at birth include breathing, beating of the heart, pulsation of the umbilical cord and definite movement of voluntary muscles whether or not the umbilical cord has been cut or the placenta is attached. Fleeting reflex activity observed only in the first minute after birth does not warrant classification as a sign of life.

Notes for national and international reporting (See also [2.25.4.5](#)):

- Definitions and reporting criteria concerning the lower limit for fetal deaths or spontaneous abortions may differ depending on different national legislation

(generally medically recognised embryonic period of gestation lasts until the 12th completed week, however another national legal limit could be set at fetus of for example 11 or 13 weeks). Any lower limit should be specified in the statistics produced.

- It is recommended to produce statistics for all fetal deaths and all deaths following live births, while reporting criteria may differ depending on different national legislation.
- For international reporting it is recommended to report stillbirths of 28 or more completed weeks of gestation (late stillbirth) and all deaths following live birth. Countries with ability for reporting stillbirths of 22 or more completed weeks of gestation (early stillbirth) are recommended to do so.
- When information on gestational age is unavailable for spontaneous abortion or stillbirth, use birthweight less than 500 grams as the criteria.

Artificial termination of pregnancy

- **Artificial termination of pregnancy** is a complete expulsion or extraction from a woman of an embryo or a fetus (irrespective of the duration of the pregnancy), following a deliberate interruption of an ongoing pregnancy by medical or surgical means, which is not intended to result in a live birth.

Artificial termination of an ongoing pregnancy is regulated by law and may be referred to as either legal abortion, induced abortion, fetal reduction or other terminologies. As long as it meets the definition of artificial termination of pregnancy it should be considered separately from spontaneous abortion or stillbirth and clearly distinguishable in the statistics.

Whilst transient signs of life may be evident after some cases of artificial termination of pregnancy at later gestational weeks, these should never be coded as spontaneous abortion or live births.

2.25.4.3 Other terminologies used in recording and presentation of perinatal or neonatal mortality

Fetal death (i.e. regardless of gestational age; lower limit, if any, should be stated)

- **Antepartum fetal death** is a fetal death before the onset of labour. If vital status of the fetus at the onset of labour is unknown, consider it was antepartum if there is presence of signs of maceration at the time of delivery.
- **Intrapartum fetal death** is a fetal death during labour. If vital status of the fetus at the onset of labour is unknown, consider it was intrapartum if there is fresh skin appearance or no signs of maceration at the time of delivery.

Stillbirth (i.e. 22 or more completed weeks)

- **Early stillbirth** is stillbirth of 22-27 completed weeks of gestation.
- **Late stillbirth** is stillbirth of 28 or more completed weeks of gestation.
- **Antepartum stillbirth** is stillbirth following antepartum fetal death (i.e. occurring before onset of labour).
- **Intrapartum stillbirth** is stillbirth following intrapartum fetal death (i.e. occurring during labour).
- **Macerated stillbirth** is stillbirth with presence of signs of maceration at the time of delivery.
- **Fresh stillbirth** is stillbirth with fresh skin appearance and no signs of maceration at the time of delivery.

Period of gestation

- **Preterm** is less than 37 completed weeks (less than 259 days) of gestation.
- **Term** is from 37 completed weeks to less than 42 completed weeks (259 to 293 days) of gestation.
- **Post-term** is 42 completed weeks or more (294 days or more) of gestation.

Birthweight

- **Extremely low birthweight** is a birthweight less than 1000 g (up to and including 999 g).
- **Very low birthweight** is a birthweight less than 1500 g (up to and including 1499 g).
- **Low birthweight** is a birthweight less than 2500 g (up to and including 2499 g).
- **Large birthweight** is a birthweight of 4000 g or more and less than 4500 g (up to and including 4499 g).
- **Exceptionally large birthweight** is a birthweight of 4500 g or more.

Neonatal death

- **An early neonatal death** is a death during the first 7 completed days after live birth (days 0 - 6)
- **A neonatal death** is a death during the first 28 completed days after live birth (days 0-27)

Total birth

- **Total birth** is the total of stillbirth and live birth. If the lower limit of stillbirth is set different from 22 weeks, for example at 28 weeks for international reporting, this should be clearly stated.

2.25.4.4 Certification of stillbirth and live births in the neonatal period

The reliability of the mortality estimates related to children depends on accuracy and completeness of recording and reporting of births and deaths. A death certificate should be issued for all fetal deaths regardless of its occurrence being spontaneous or deliberate, and for all neonatal deaths following live births.

Under-reporting and misclassification are common, especially for stillbirths and early neonatal deaths. Countries should arrange registration and reporting procedures so that the events and the criteria for their inclusion in the statistics can be easily identified.

Data of fetal deaths meeting requirements for stillbirth registration should be included in mortality statistics, but should be clearly distinguishable from neonatal deaths following live births. Follow instructions in [2.25.4.5](#) for details, when it is not clear whether the certificate relates to a stillbirth or a neonatal death following live birth.

Artificial termination of pregnancy should also be included in mortality statistics, but should be clearly distinguishable from spontaneous abortion or stillbirths.

The international form of medical certificate of cause of death and additional details

With the update of the International form of medical certificate of cause of death in 2016, just one certificate is used for all cases including stillbirths (see [2.15](#)). Care needs to be taken to correctly fill in the specific section for perinatal deaths on the certificate. The previously recommended perinatal death certificate should be replaced by the form in [2.15](#), and perinatal deaths should be coded according to the general mortality coding instructions.

While not required for ICD-11 coding, the additional information mentioned in [2.15](#) might be helpful for the monitoring of perinatal and infant deaths of a country or region.

Level of details for recording

Gestational age and birthweight should be recorded for all live births and stillbirths to the degree of accuracy to which it is measured.

The gestational age should be recorded in number of completed weeks as a minimum, and where possible an additional variable capturing the number of days should also be included.

While statistical tabulations include 500 g groupings for birthweight, weights should not be recorded in those groupings. The actual weight should be recorded to the degree of accuracy to which it is measured.

Chronological age at death during the first 24 hours of life (day 0) should be recorded in units of completed minutes or hours of life. For the second (day 1), third (day 2) and through 27 completed days of life, age at death should be recorded in days. It is recommended that these data are collected as part of a minimal perinatal dataset.

2.25.4.5 Reporting criteria for fetal death, stillbirth and live birth

As a minimum all stillbirths and deaths following live births born with 22 or more completed weeks of gestation (≥ 154 days) should be included in the statistics, though the legal requirements for the registration may vary depending on different national legislations.

When information on gestational age is not available, the corresponding criteria for birthweight (500g or more) should be used.

Where gestational age and birthweight are not known, for example when only crown-heel length or estimated gestational age based on physical examination is available, or when no such information is available, the event should be included in, rather than excluded from, mortality statistics of the perinatal period.

Where fetal deaths and/or neonatal deaths at <22 completed weeks of gestation (or <500g) are included in perinatal statistics these should be presented separately from stillbirths and/or neonatal deaths at 22 or more completed weeks of gestation and the lower limit for inclusion in perinatal statistics in the setting should be stated, for example '20 completed weeks of gestation' or 'no lower gestational age limit'.

Note that the reporting criteria above do not apply to artificial termination of pregnancy, which is defined irrespective of duration of pregnancy and should be presented separately from fetal death, stillbirth or live birth.

Criteria for international reporting

In statistics for international comparison, inclusion of fetal deaths and live births born at extremely low gestational ages disrupts the validity of comparisons and is not recommended. Therefore, the criteria for international reporting of stillbirths and/or neonatal deaths are set at 28 completed weeks or more, while countries are encouraged to provide, where possible, data from 22 completed weeks or more as well.

Table 1

Gestational age (completed weeks)	Birthweight (grams)	Recommendations for registration of data
<22 (very early gestation)	<500	Only required for neonatal mortality national statistics. When used for spontaneous abortion (miscarriages), should be reported separately from perinatal statistics and the lower gestational age limit for data collection stated.
22-27 (early gestation)	500 or more	For national statistics, and for international statistics of countries with ability for reporting of early gestation deaths (stillbirths and neonatal mortality)
28 or more (late gestation)	1000 or more	For international statistics (stillbirths and neonatal mortality)
Unknown gestational age	Unknown birthweight	Include in statistics only when there is a high likelihood that the stillbirth or neonatal death occurred at the given criteria e.g. 28 or more weeks for international statistics.

2.25.4.6 Statistical presentation of perinatal, neonatal, infant or under-five mortality

For perinatal mortality statistics, full-scale multiple- cause analysis of all conditions reported will be of the greatest value.

Countries should provide the rates listed below for international comparisons:

- **Late stillbirth rate** = (stillbirths $\geq$ 28 weeks gestation/total births (stillbirths $\geq$ 28 weeks gestation and live births) x 1000
- **Early neonatal mortality rate** = (early neonatal deaths (day 0-6) $\geq$ 28 weeks gestation /live births $\geq$ 28 weeks gestation) x 1000
- **Perinatal mortality rate** = (stillbirths $\geq$ 28 weeks gestation and early neonatal deaths (day 0-6) $\geq$ 28 weeks gestation) /total births (stillbirths $\geq$ 28 weeks gestation and live births) x 1000

Groupings of gestational age groups for fetal death under 22 weeks, stillbirth and neonatal mortality statistics

<22 completed weeks (<154 days) N.B. where the under 22 week category is used, the lower limit included should be specified

22 - 27 completed weeks (154 - <196 days)

28 - 31 completed weeks (196 - 223 days)

32 - 36 completed weeks (224 - 258 days)

37 - 41 completed weeks (259 - 293 days)

42 completed weeks and over (294 days and over)

Groupings of birthweight for fetal death under 22 weeks, stillbirth and neonatal mortality statistics

By weight intervals of 500 grams, i.e. 1000-1499 grams, etc.

499 grams or less 500 – 999 grams 1000 - 1499 grams 1500 - 1999 grams 2000 - 2499 grams
2500 - 2999 grams 3000 - 3499 grams 3500 - 3999 grams 4000 - 4499 grams 4500 - 4999
grams 5000 grams or more

Groupings by chronological age in neonatal mortality statistics

- *Preferred groupings:*
By single days for the first week of life (under 24 hours (day 0), 1, 2, 3, 4, 5, 6 days), 7-13 days, 14- 20 days, 21-27 days
- *Alternatives if preferred groupings is not available:*
Under 24 hours, 1-6 days, 7-27 days,
Under 7 days, 7-27 days
- *Additional chronological age groupings for day 0 neonatal deaths:*
Under 1 hour, 1–11 hours, 12–23 hours

2.25.4.7 Under-five mortality

Under-5 mortality (Under-5 mortality rate – probability of dying between birth and exactly 5 years of age, expressed per 1,000 live births.) is a leading indicator of the level of child health, quality of life, health infrastructure, and overall development in countries. It is also the SDG indicator.

2.25.4.8 Infant mortality

Infant mortality (infant mortality rate – probability of dying between birth and exactly 1 year of age, expressed per 1,000 live births) is an indicator for quality of life and health infrastructure.

2.25.5 Standards and reporting requirements related for maternal mortality

Maternal mortality is part of the assessment of the Sustainable Development Goals (SDG) that serve to monitor the impact of the joint work of the international community in this field.

2.25.5.1 Maternal death

A maternal death is defined as the death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the duration and site of the pregnancy, from any cause related to or aggravated by the pregnancy or its management, but not from accidental or incidental causes.

2.25.5.2 Late Maternal death

A late maternal death is defined as: the death of a woman from direct or indirect obstetric causes, more than 42 days but less than one year after termination of pregnancy.

2.25.5.3 Comprehensive maternal death

A grouping that combines maternal death and late maternal death.

2.25.5.4 Direct and indirect obstetric deaths

Maternal deaths, late maternal deaths, and comprehensive maternal deaths are subdivided into two groups:

- Direct obstetric deaths: those resulting from obstetric complications of the pregnant state (pregnancy, labour, and puerperium), and from interventions, omissions, incorrect treatment, or from a chain of events resulting from any of the above.
- Indirect obstetric deaths: those resulting from previously existing disease or disease that developed during pregnancy, and that were not due to direct obstetric causes but were aggravated by the physiologic effects of pregnancy.

2.25.5.5 Death occurring during pregnancy, childbirth and puerperium

A death occurring during pregnancy, childbirth, and puerperium is defined as: the death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the cause of death (direct and indirect obstetric and non-obstetric death).

2.25.5.6 Recording requirements of maternal mortality

In order to improve the quality of maternal mortality data and provide alternative methods of collecting data on deaths during pregnancy or related to pregnancy, as well as to encourage the recording of deaths from obstetric causes occurring more than 42 days following termination of pregnancy, the Forty-third World Health Assembly in 1990 adopted the recommendation that countries consider including questions regarding current pregnancy and pregnancy within one year preceding death on death certificates.

The classification also allows the recording of deaths that occur one year or more after termination of the pregnancy ([JB62](#) Death from sequelae of obstetric causes).

2.25.5.7 International reporting of maternal mortality

For the purpose of international reporting of maternal mortality, only those maternal deaths occurring before the end of the 42-day reference period should be included in the calculation of the various ratios and rates, although the recording of later deaths is useful for national analytical purposes.

2.25.5.8 Numerator, denominator, and ratios of published maternal mortality

Published maternal mortality ratios should always specify the numerator, which can be given as the number of recorded direct obstetric deaths, or the number of recorded obstetric deaths (direct plus indirect). Note that cases not coded to Chapter 18 should also be included in the numerator. These include those categories presented in the 'Exclusion Note' at the beginning of Chapter 18, provided that they have been aggravated by pregnancy or conversely aggravated the pregnancy.